

MobileTinyVit: Rapid and Interpretable Detection of Cotton Leaf and Plant Diseases Using Lightweight Transformer Ensemble

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ABSTRACT

Cotton leaf and plant diseases pose a big challenge to crop quantity and quality necessitating timely and precise detection of the condition as a means of maintaining sustainable agricultural crops. The conventional methods of identifying diseases are time-consuming, labor-intensive, and prone to human error. Latest developments with deep learning and transformer-based models provide automated, robust, and fast methods to detect and identify disease in leaves and entire plants with high accuracy. Nonetheless, most of the current methods have weaknesses such as lack of rigorous evaluation, either leaf or plant-only approaches, and are not interpretable, which are reasons that prevent their practical implementation in the real world. To overcome these limitations, this research paper introduces MobileTinyVit, a lightweight soft voting ensemble that fuses MobileViT S and TinyViT 5M 224. The model has a high level of accuracy because it uses the complementary strengths in local and global feature extraction and is computationally efficient. Through extensive preprocessing such as resizing, histogram equalization, Gaussian blurring, and normalization, the model was robust and data augmentation increased generalization as demonstrated by an accuracy of 99.57%, F1 of 1.0000, Cohen kappa of 0.9940, and consistency of cross-validation with 5-fold cross-validation of 0.9974 ± 0.0007 . The proposed model takes a total inference time of approximately 0.56s using a small amount of GPU. Gradient-weighted Class Activation Mapping (GRAD-Cam) visualizations confirm that the model is correct in terms of targeting disease-specific areas, which enhances interpretability. We also create a user-friendly web application that can generate real-time predictions within less than 2 seconds. The reported F1-score of 1.0000 is the weighted macro-average across the classes and the overall accuracy of 99.57% shows minor misclassifications for rare edge cases. These values are perfectly in line with the ones extracted from the confusion matrix and 5-fold cross-validation (mean accuracy $99.74\% \pm 0.07\%$), also confirming that the model delivers robust near-perfect performance, which is indicative the viability of the MobileTinyViT as a practical application in precision agriculture and fast disease control

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1. INTRODUCTION

Cotton leaf ailments pose a great limitation to the worldwide yield of cotton causing a considerable decline in yield and financial losses to the farmers [1]. These diseases need to be diagnosed correctly and early, so that management and prevention of these diseases will be effective and that the time taken in reducing the use of chemicals will be in time. Conventional diagnostic systems are based on visually examining which is labor intensive, subjective and is likely to create inconsistencies especially in the large scale farming processes [2–4]. These conventional methods most often involve expert visual evaluation based on symptoms by bacterial blight spots, fungal lesions (e.g., *Alternaria* leaf spot), nutrient deficiencies (yellowing), or pest damage, together with field charts of symptoms, physical sampling, and the use of laboratory tests for confirmation.

The development of computer vision and deep learning has transformed the idea of plant disease detection, and convolutional neural networks (CNNs) and transformer-based models demonstrate impressive results in the context of automated classification [5, 6]. A number of studies have shown that CNNs, hybrid, and attention-based frameworks are highly accurate in detecting cotton leaf disease. Nonetheless, most of these methods have limitations in the form of small and unbalanced samples, cross field no longer, and overfitting to laboratory conditions [7, 8]. Moreover, the vast majority of currently available models are computationally intensive, and it is difficult to deploy them in resource-constrained devices, like mobile devices or edge systems [9].

Moreover, existing literature tends to pay attention to single-model structures that might not represent the entire range of discriminative characteristics of complicated disease patterns [10–14]. The interpretability of the models is also not given much consideration, which minimizes the trust on the decisions made in the real world. Models with a balance between predictive accuracy, computational efficiency, and robustness to a wide variety of conditions in the field have not been studied much. All these issues highlight the necessity of a new strategy that incorporates the advantages of successful models and at the same time should be efficient and applicable. To overcome the shortcomings of existing methods for cotton leaf disease detection, we introduce *MobileTinyViT*, a solution that combines preprocessing and data augmentation with lightweight transformer-based structures. This combination of *MobileViT* and *TinyViT* models is a trade-off of both computation and prediction performance, and thus, it is best to be deployed in resource-constrained systems, including mobile devices and edge systems. Our 5-fold cross-validation is used to ensure strong performance, maximize generalizability and minimize overfitting at minimal computational cost. We employ Grad-CAM in order to make the predictions of the models easier to interpret so that the users can see the main areas of the leaf that are considered by the model. This openness enhances the trust of the users and the decision-making. Moreover, we conduct extensive statistical analysis with the help of such measures as accuracy, precision, recall, F1-score, and stability tests and conclude on the performance of the models. Lastly, we create a simple demo application, that works well with low resource devices, which enables farmers to send images of leaves and get real-time feedback on the diseases detected. The purpose of this tool would be to offer actionable information promptly to enhance the management of the cotton crop, as well as to encourage sustainable agricultural practices. So, we can summarize the contribution of this study as follows:

1. This study proposes *MobileTinyVit*, a lightweight soft-voting ensemble that integrates *MobileViT-S* and *TinyViT-5M-224* for accurate cotton leaf and plant disease classification.
2. The proposed model achieves state-of-the-art performance with 99.57% accuracy, a perfect F1-score, and stable 5-fold cross-validation results, demonstrating strong generalization capability.
3. Grad-CAM-based interpretability is incorporated to highlight disease-specific regions, enhancing transparency and trust in the model's decision-making process.
4. A real-time web application is developed to enable fast and user-friendly disease prediction, demonstrating the practical applicability of the proposed system in precision agriculture.

In conclusion, the paper introduces *MobileTinyViT*, a lightweight ensemble of *MobileViT* and *TinyViT*, which relies on effective preprocessing, data augmentation, and efficient transformer designs to predict cotton leaf and plant diseases in an effective and efficient way. The proposed framework is highly performing and has high transparency with the inclusion of Grad-CAM-based interpretability and 5-fold cross-validation, making it robust and appropriate to implement in real-world settings. Moreover, a working web-based demonstration program is designed to support real-time disease detection to provide farmers with actionable information on time to assist in better crop management and decision-making.

The rest of the paper will be structured as follows: Section 2 will review the related works on cotton leaf disease detection, Section 3 will detail the proposed methodology, Section 4 will present and discuss the experimental results and Section 5 will conclude the paper and outline future research directions.

2. REVIEW OF RECENT WORKS ON COTTON AND PLANT LEAF DISEASE

Cotton and plant leaf disease detection has received considerable attention over the last few years because it directly affects agricultural productivity and sustainability. As the field of deep learning and computer vision advances, scholars have suggested many architectures and hybrid frameworks to enhance the detection accuracies, efficiency, and interpretability. The section contains the review of recent state-of-the-art based on CNNs, transformers, and explainable AI to classify cotton and plant leaves in the presence of diseases and its methods, performance, and limitations. Rehman et al. [15] proposed a hybrid formal verification and deep-learning framework, combining TLA+ with CNN/LSTM/RNN to detect cotton leaf diseases. Their CNN achieved 98.7% accuracy with high precision (96.8%) and recall (98.0%), along with F1-scores and AUC, highlighting the superiority of CNN for fine-grained plant-state data. However, the method's reliance on high-quality images (requiring powerful hardware, such as ResNet152-style models) and potential overfitting to controlled datasets are notable limitations. Real-field conditions, including variations in lighting and background, may degrade the model's performance.

Kaur et al. [16] developed an explainable deep model combining EfficientNetB3 and InceptionResNetV2, optimized using a genetic algorithm, for cotton leaf disease classification. Their hybrid model, which integrated XAI techniques such as LIME and SHAP, reached 98.0% accuracy ($F1 \approx 98.0\%$, $AUC \approx 0.9992$). This lightweight framework is suitable for mobile and IoT deployments. However, the training was based on a relatively uniform Kaggle dataset, and the model's generalization to diverse real-world conditions remains untested. The authors also highlighted the need for further work on XAI methods and real-time deployment challenges.

Aslam et al. [8] proposed "StackNet," an ensemble of CNN feature extractors and machine learning classifiers to detect cotton leaf diseases. The model utilized synthetic image generation via GAN to address class imbalance and leveraged pre-trained MobileNet/VGG16 features with LSTM, SVM, and RF classifiers before stacking. The ensemble achieved approximately 97% accuracy on a 7-class dataset, outperforming individual models. However, GAN synthesis may introduce artifacts, and the model's interpretability is limited. Furthermore, training multiple networks increases computational costs and the risk of overfitting, especially if synthetic and real data distributions diverge.

Islam et al. [10] evaluated fine-tuned CNN architectures, including VGG16, VGG19, Inception-V3, and Xception, for cotton disease classification. The Xception model achieved the highest accuracy of 98.70%. The study suggests that deeper architectures are better at extracting features from leaf images. However, class imbalance and the lack of explicit feature segmentation were observed as issues. The reliance on a Kaggle dataset that may not reflect field variability and the need for integrating ROI segmentation or hybrid models are noted as key limitations. Additionally, the system's need for IoT/web integration for real-time use is emphasized.

Singh et al. [17] introduced BERT-ResNet-PSO, a hybrid model where a Vision-BERT-like encoder preprocesses image patches, ResNet extracts features, and Particle Swarm Optimization tunes the pipeline. On a large PlantVillage cotton dataset, the model achieved 98.5% accuracy (Precision 98.2%, Recall 98.7%, F1 98.4%). Compared to baseline ResNet, the inclusion of BERT improved feature representation. However, the increased complexity of the BERT encoder and the need for validation on unseen varieties/environments present significant challenges. The model's computational cost and the need for more robust training, particularly for rare classes, were also identified as limitations.

Faisal et al. [14] benchmarked multiple CNNs, including VGG16, DenseNet, EfficientNet, InceptionV3, MobileNet, NASNet, and ResNet, on a real-field cotton disease dataset. The ResNet152-based model achieved the highest performance with approximately 99% accuracy. However, ResNet152 is computationally heavy and impractical for growers with limited resources. The data used for training lacked challenging real-world conditions such as low light and occlusion, and the robustness of the model to changes in image resolution remains untested. These limitations suggest that a lighter model or pruning may be required for real-world deployment.

Akbar et al. [18] addressed cotton leaf lesion detection by combining CNN backbones with data aug-

mentation. They fused features from VGG16, InceptionV3, and ResNet50 and applied a Wasserstein GAN to generate synthetic images for underrepresented classes. Their model achieved 95% accuracy and 98% F1-score on a 4-class dataset. GAN augmentation helped mitigate class imbalance, but GAN-generated images may not fully capture lesion variability. Additionally, the small evaluation dataset raises concerns about overfitting. The ensemble approach also increases inference time and resource needs, which may hinder deployment on mobile devices.

Joshi et al. [19] implemented YOLOv8, an object-detection/classification model, with 10-fold cross-validation to classify cotton leaf diseases. They achieved a Top-1 accuracy of 99.60% with $F1 \approx 99.6\%$, recall/precision $\approx 99.5\%$ on a 7-class dataset. However, their experiments were based on a controlled dataset, and YOLOv8 may misclassify in unstructured field images. Moreover, the increased inference cost compared to simpler CNN models could limit mobile use. The authors suggest external validation on unseen farms is needed to assess generalizability.

Ahmad et al. [12] experimented with Vision Transformers (ViT) and CNNs on an Indian cotton disease dataset. The ViT models achieved 96.72% accuracy on binary classification (healthy vs diseased) and 93.39% on multi-class classification. While ViT models captured global leaf context effectively, they underperformed CNNs in multiclass classification, suggesting that transformers may require more data or tuning to handle fine-grained texture differences. The study also lacked model explainability, and future work should integrate attention visualization to improve interpretability.

Palve et al. [20] compared traditional machine learning classifiers (SVM, KNN, NB, DT, shallow NN) on a 4-class cotton leaf dataset. Their results showed that Neural Networks and SVM achieved the highest accuracy (96%), while KNN performed poorly at 65%. Although the study confirms that SVM/NN can match CNN performance on limited datasets, the absence of deep CNN features and the limited Kaggle dataset were identified as drawbacks. The study also suggested that CNNs would outperform traditional ML methods on larger and more complex datasets.

Nachankar [21] proposed a transfer-learning-based CNN, achieving 96.4% accuracy for diseased vs. fresh classification. The study emphasized that transfer learning significantly boosts performance, but the classes were broad and not fine disease types. The limited dataset size (2400 images) raised concerns, and scaling to more disease categories or real-field images could affect accuracy.

Shao et al. [13] introduced CANnet, a custom CNN with novel modules like Receptive Field Space Channel RFSC and Coordinate Attention PCA, reaching 96.3% accuracy on a self-collected cotton disease dataset and 98.6% on a public dataset. Their model effectively captures both local and global leaf features. However, the complex modules add many parameters, increasing the risk of overfitting. Generalization to different lighting conditions and backgrounds is unproven.

Memon et al. [22] proposed a “meta deep learning” model, which outperformed others with 98.53% accuracy on the cotton leaf dataset. While the study demonstrated that meta-learning can enhance classification, the small dataset size and the lack of model interpretability are notable limitations. Future work should test the model on larger, more varied datasets and compare it with more recent architectures.

Summary:

The current literature on the detection of cotton leaf disease has come a long way, however, there are a number of critical gaps and constraints within the literature that prevents their practical implementation in the real world. Most of the models rely on the quality and well-centered images and therefore fail to generalize to the real world situation where the image can be affected by variations in lighting, backgrounds, and occlusions. Moreover, although the methods involving the use of GAN-based augmentation are mentioned in some works as the solution to the issue of an imbalanced class, these approaches introduce artifacts, and the synthetic data might not be as diverse as real lesions, which might cause excessive overfitting. A second significant drawback is that most models have not been generalized to field conditions with a large number having been evaluated on controlled datasets and not validated in a variety of environmental conditions, including varying lighting or backgrounds, which can lead to poor performance. Another important obstacle is the high computational expense of deep learning models especially the ones based on architectures such as ResNet152, ViT, and BERT. Such models demand a strong hardware, which is why they cannot be deployed to devices with limited resources, like mobile phones or IoT systems, which are essential to on-the-ground operations of farmers. Although some of explainable AI methods such as LIME, SHAP and Grad-Cam being investigated in some works, model interpretability is the bottleneck in many existing methods because farmers and agriculture experts require easily understandable illustration of the models' decisions (e.g., disease-affected patches on

leaves) to trust comfortably for implementing focused treatments successfully, to prevent unwarranted uses of pesticides and to make smart decisions regarding crop management during in-field operations[6, 7, 9, 16]. Because of this lack of interpretability, black-box predictions are a barrier to practical use as users cannot check that the model is indeed attending to disease symptoms and not irrelevant background features (soil or lighting artifacts)[3, 23].

A lack of transparency in the decision making process may also lead to decreased user trust and impractical adoption. Moreover, there are still problems with deployment in real-time, with most models not being optimized in terms of low-latency inference or consistent functionality in field settings. Small and low-diversity datasets used in various studies also increase the risk of overfitting which limits generalization to unseen conditions. Repeat such limitations, such as overfitting with small datasets, poor field generalization and high computational requirements in hybrid models also are frequently discussed across the most recent papers highlighting systemic deficiencies of AI in agriculture practice rather than sporadic one [2, 3, 6, 9].

Our approach of lightweight ensemble design by solving multiple overlapped challenges at a time helps us overcome the repetitive limitations found in single-model/huge hybrid systems. Although the hybrid models have good performances, they can be computationally expensive and hard to implement especially in limited resource-based agricultural platforms. These limitations are necessary to develop the strong, scalable, and interpretable disease detection systems that can be integrated successfully into the real-life agricultural activities and assist farmers in a timely and precise manner.

3. METHODOLOGY

This research is conducted systematically to design a precise and effective cotton plant and leaf disease diagnosis system. This research approach starts with data acquisition, and the data acquired should be comprehensive and representative. This is followed by preprocessing, augmentation, and feature extraction methods that are used to improve model training. Lastly, a slim ensemble model is trained, validated, and tested to prove that it is not only high-performing but can also be deployed practically. Figure 1 shows the overall workflow diagram of the proposed cotton leaf disease diagnosis framework.

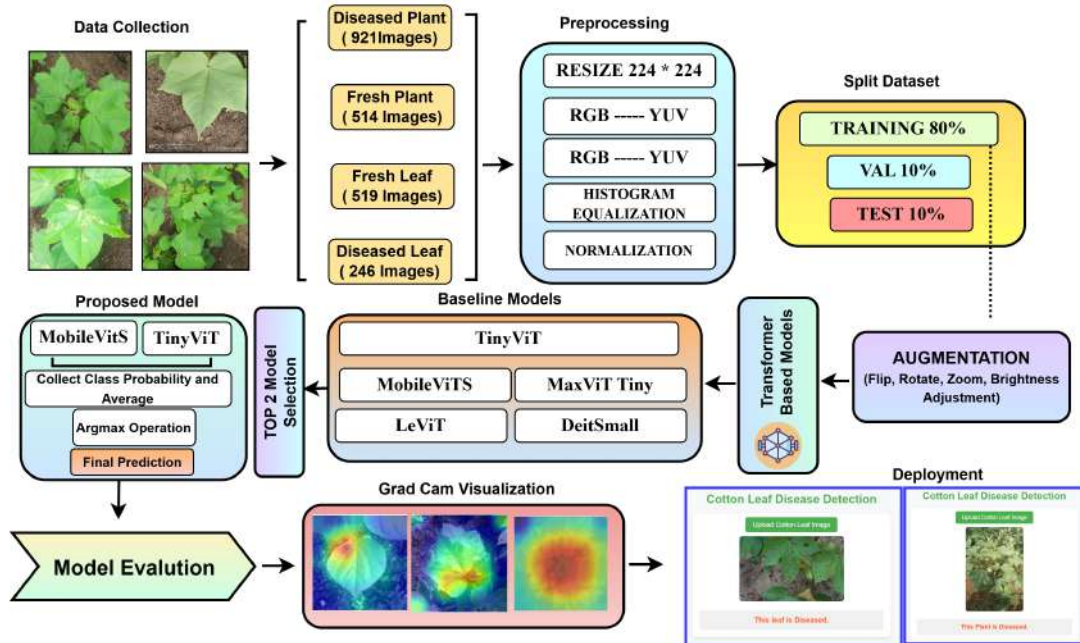


Figure 1. Overall workflow of the proposed cotton leaf and plant disease diagnosis framework, illustrating data acquisition, preprocessing, augmentation, feature extraction, and ensemble model training and evaluation.

3.1. Data Acquisition

The data utilized in the present research was gathered from Kaggle and consists of pictures of cotton plants and leaves that were taken in the natural environment and in the laboratory [24]. The data set includes

four different classes: diseased cotton leaf, diseased cotton plant, fresh cotton leaf and fresh cotton plant, and it includes 2300 images. The total number of images in each category is presented in Table 1.

Table 1. Overview of the dataset used in this study.

Category	Description	Image Count
Diseased Cotton Leaf	Images of cotton leaves showing signs of disease.	346
Diseased Cotton Plant	Images of whole cotton plants affected by disease.	921
Fresh Cotton Leaf	Images of healthy cotton leaves.	519
Fresh Cotton Plant	Images of healthy cotton plants.	514

Also, figure 2 shows a sample from the dataset, illustrating various examples of cotton plants and leaves in both diseased and healthy states.



Figure 2. A sample from the dataset showing both diseased and healthy cotton leaves and plants.

3.2. Preprocessing

The preprocessing of the images is vital in improving the quality of the input data and producing uniform input when training the model. In this study, the pipeline is used with four primary steps: resizing, histogram equalization, Gaussian blurring, and normalization. To begin with, each image is downsampled to a constant size in order to create inputs of the same size as required by deep learning models, in particular, convolutional neural networks. Resizing is used to standardize the input without losing valuable information and adding an unnecessary complication. This YUV transformation eliminates this unnecessary difficulty: direct

histogram equalization on RGB may lead to unnatural color changes that twist disease symptoms (e.g., yellowing or spot). By normalizing only the Y (luminance) component, we enhance contrast in lesions consistently without changing original colors.

The resized images are then translated out of the RGB color space into the YUV. This color space is a separation of the luminance (Y) and chrominance (UV). Histogram equalization is done in the Y (luminance) channel to increase contrast so that features can be seen in bright and dark areas of the image. A 3x3 kernel is applied to smooth the image with the help of Gaussian blurring to remove noise. This assists in reducing high frequency noise to enable the model to be concentrated on significant patterns. Lastly, the pixel values are scaled to the range of 0 to 1 by converting it into floating-point values and then dividing by 255. This standardization keeps the features of input in a similar scale stabilizing and speeding up the training process.

The preprocessing pipeline thus improves image quality and prepares the data for effective model training.

Figure 3 illustrates a sample image after all preprocessing steps, showing the improvements in quality and consistency.



Figure 3. A sample preprocessed image after resizing, histogram equalization, Gaussian blurring, and normalization.

3.3. Splitting Dataset

This study utilized a dataset that was further subdivided into training, validation and test sets. The data were divided with a standard 80-10-10 ratio containing 80% of the data in the training set, 10% in the validation set, and 10% in the test set. By doing so, the model will be trained on a large fraction of the data, but still have the validation and test sets that are large enough to evaluate the generalization capacity of the model.

Figure 4 illustrates the distribution of the data across these three sets, providing a visual representation of the 80-10-10 split.

3.4. Augmenting the Training Set to Increase Diversity

In order to increase generalization of the model, several types of image augmentation methods were used in the training data. These were random flipping, rotating, zooming and brightness settings. Random flipping was used to give the impression of variations in object orientation, whereas random rotations (between -25 to 25) enabled the object to be invariant to small orientation variations. The zoom effect, used between 80% and 120% of the original sizes, was used to reflect a difference in the size of the object or random changes in brightness, which explained the variation in the lighting conditions. To make the data more diverse and avoid

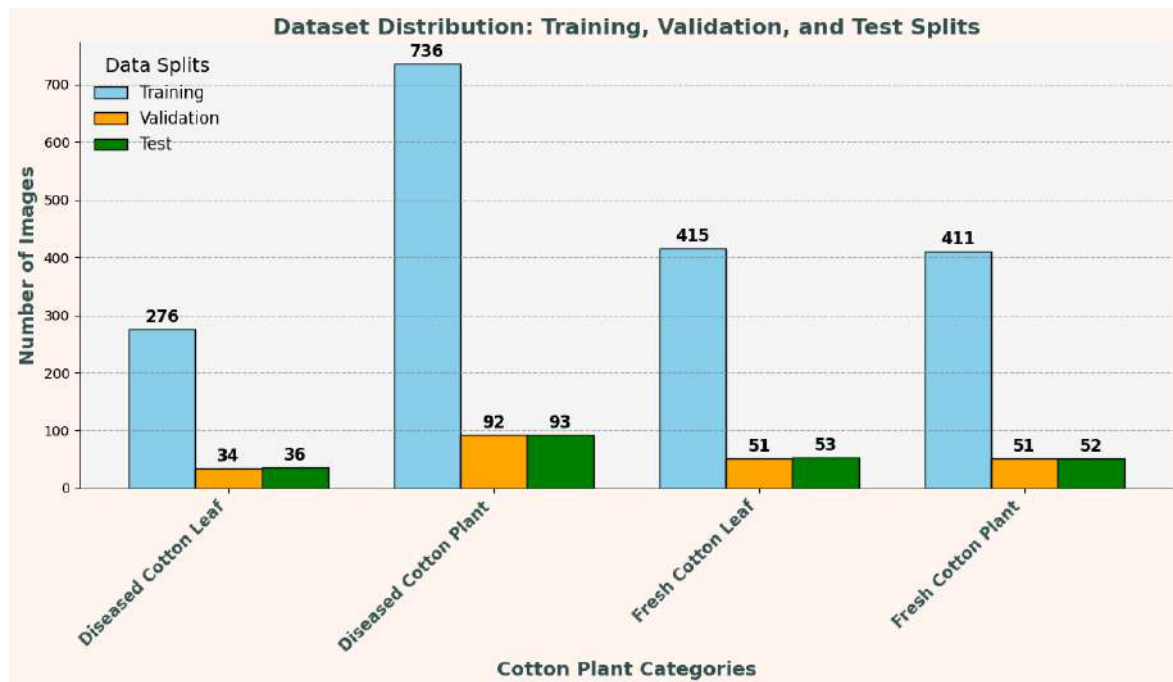


Figure 4. Distribution of the dataset into training, validation, and test sets (80-10-10 split).

overfitting, each original image was supplemented with three variations of it. The final image count per class after augmentation is shown in the table 2 below:

Table 2. Image count per class after augmentation

Class	Original Count	Augmented Count
Diseased Cotton Leaf	276	1,104
Diseased Cotton Plant	736	2,944
Fresh Cotton Leaf	415	1,660
Fresh Cotton Plant	411	1,644

This augmentation process significantly increased the training set size and variability, enabling the model to better generalize across diverse real-world conditions. As listed in table 2, the original dataset was an imbalanced one (i.e. less diseased leaf and plant samples). Over-sampling using augmentation balanced class counts (now 1100–1600 per class) and increased training data 6×. This reduced bias towards majority classes, decreased overfitting and led to better generalization of the model on unseen field images which resulted in higher accuracy.

3.5. Baseline Classifiers

This paper compares various transformer-based architectures as benchmark classifiers. These models are characterized by a spectrum of complexities and are frequently applicable in the task of image classification. Each model was chosen to illustrate a range of approaches including lightweight models based on edge devices and more robust models that aim at achieving accuracy. Although several other transformer variants exist, these models were selected due to their lightweight and compact architectures. Ensembling larger transformer models would significantly increase computational cost, memory consumption, and inference latency, making real-time deployment impractical, especially in resource-constrained environments

MobileViT-S [25]: The MobileViT-S is a small-sized vision transformer that integrates the benefits of convolutional neural networks (CNNs) with transformers to be applied in mobile and edge devices. MobileViT-S applies the hybrid architecture, i.e. CNN layers are applied to extract local features, and transformer layers are applied to extract global dependencies across the image. The model uses a small scale of the Vision Transformer (ViT) blocks, having less parameters as compared to the traditional ViT models, and is thus efficient

in memory and computation usage. It has depthwise separable convolutions in its architecture that can greatly lower the computational cost. MobileViT-S is able to achieve a competitive accuracy despite having a small number of parameters, usually 2.2 million, and working on 224x224 input images. The processing capability of the model in high-dimensional data and its computational efficiency makes it suitable in applications that use real-time in the mobile environment.

MaxViT-Tiny [26]: The model MaxViT-Tiny represents a dynamic attention pooling mechanism that allows the model to represent long-range dependencies in a more efficient manner. In contrast to the conventional ViT with fixed attention mechanism, MaxViT is a scale-adaptive tokenization module that adapts itself to the content of the picture, both to local and global features. MaxViT-Tiny is a smaller model, which has only 2.9 million parameters, is therefore highly effective in mobile and embedded systems. The design reflects multi-scale attention blocks which can be run at varying resolutions thereby improving the concentration of the model on the image elements of relevance. An input size of 224x224 is provided by MaxViT-Tiny and this model is specifically useful in finer grained features of high resolution images owing to the special pooling mechanism. The attention layers dynamically fuse image patches according to their spatial significance and thereby, saves on computation and preserves significant image details.

LeViT-192 [27]: The LeViT-192 model is a part of the LeViT family that aims at the attainment of a small model with an effective attention mechanism. The LeViT is a hybrid architecture, which uses convolutional layers with the transformer layers to design an effective network to carry out the image classification tasks. The model has 12 transformer layers with multi-head self-attention of the model which enables it to generate both short and long dependency in images. The attention heads in LeViT are made smaller, and the model has a lighter variant of self-attention that makes it lighter than the traditional transformers. LeViT-192 is an input resolution of 192x192 pixels with approximately 8 million parameters. Linear attention layers are one of the most important advances of the model that allow decreasing the cost of the computations caused by the classical quadratic attention mechanism. This renders LeViT-192 very efficient but still competitive with regard to image classification tasks on benchmarks.

DeiT-Small (Patch16-224) [28]: The DeiT-Small model is a smaller variant of the Data-efficient Image Transformer (DeiT) series, designed to address the efficiency challenges associated with training vision transformers. The model is trained using a patch of 16x16 and input size of 224x224 which minimizes the dimension of the input making it train faster. DeiT-Small also includes a special teacher-student training model in which the bigger teacher model directs the training of the smaller student model, which is more effective. The architecture consists of 6 transformer layers which have multi-head self-attention, and the attention heads in every layer are 12. The number of parameters of DeiT-Small is 22 million, which is much less than larger ViT models, but the performance is competitive, and the large-scale data is not required. The most important benefit of DeiT-Small is that it can attain state-of-the-art performance using less training data and less computing resources, which is suitable in situations where there is little training data and computer resources.

TinyViT-5M-224 [29]: The *TinyViT-5M-224* model is a smaller, highly efficient hierarchical vision transformer, designed with approximately 5.4 million parameters (5M). Although it is relatively small, TinyViT still has the main strengths of the ViT architecture, as it relies on self-attention to model global dependencies, but it is structured in a hierarchical manner akin to Swin Transformer to be more efficient. It consists of four block stages of architecture with 10 Transformer layers and 2 initial convolutional (MBConv) blocks, resulting in a total of 12 blocks. It is also specialized to execute on smaller computation devices, frequently on edge devices, and provides high classification accuracy. The model uses 224x224 pixel inputs and is specifically suitable in the situation when the speed and accuracy are both important, and the balance between them is quite good.

3.6. Proposed MobileTinyVit: Soft Voting Ensemble for Cotton Leaf Disease Classification

We propose a soft-voting ensemble of MobileViT-S and TinyViT, referred to as **MobileTinyVit**, a lightweight ensemble framework for cotton leaf disease classification, as both backbone models are compact transformer architectures with reduced parameter counts, low memory consumption, and efficient inference, ensuring that the ensemble remains computationally efficient and suitable for real-time deployment. MobileTinyVit is the architecture of transformer based model which incorporates attention mechanisms of MobileViT-S and TinyViT to perform global and local feature extraction effectively for cotton leaf disease classification. The framework takes the combination of two models that use transformers, namely, MobileViT-S and Tiny-ViT-5M-224, which were chosen because they hold complementary characteristics in terms of the ability to capture local and global features. The models are independently trained to extract feature represen-

tations of the input images and the results are transformed to class probabilities through the use of a softmax layer. The ensemble combines the projections of these two best performing models through an average of the probabilities, and the ultimate class is selected as the highest combined probability. This soft method of voting offers the advantages of both of the models, enhancing their robustness and the general classification performance in comparison to the strategies of utilising a single model.

Also figure 5 shows the working architecture of the proposed MobileTinyVit.

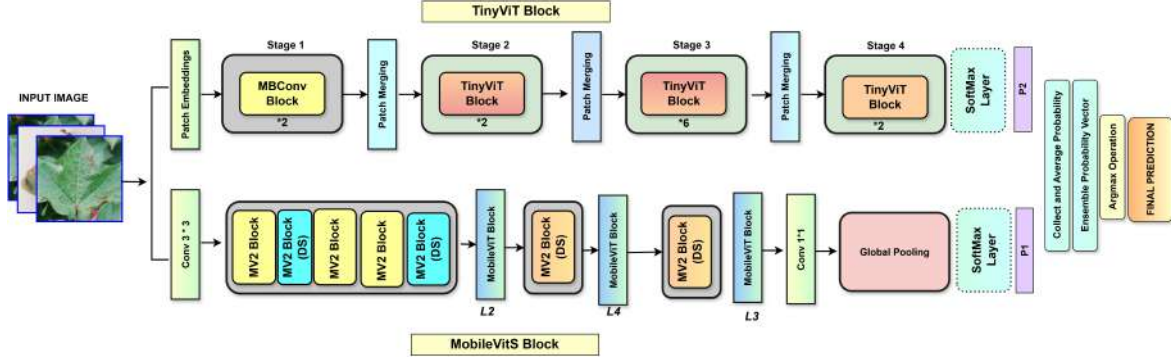


Figure 5. Architecture of the proposed MobileTinyViT-based framework.

Algorithm 3.6. shows working procedure of proposed MobileTinyViT. The MobileTinyViT is developed from soft-voting of two lightweight transformer models M_1 (MobileViT-S, a hybrid transformation with 2.2M parameters), and another pure transformer model M_2 (TinyViT-5M, a compacted final nonlocal network variant 5M). Each model feeds-forward through the input image to generate class logits. The logits are averaged and the class with maximum average probability is chosen by the ensemble.

[H] **Input:** Test dataset $\mathcal{X} = \{x_1, x_2, \dots, x_N\}$, trained models M_1 (MobileViT-S) and M_2 (TinyViT-5M-224)

Output: Predicted class labels $\hat{\mathcal{Y}} = \{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_N\}$

[1] **Motivation:** Select the two top-performing models based on validation accuracy to leverage complementary strengths and reduce model-specific errors. each test image $x_i \in \mathcal{X}$ Obtain predicted logits from each model:

$$z_1 = M_1(x_i), \quad z_2 = M_2(x_i)$$

Convert logits to class probabilities using the softmax function:

$$p_1 = \text{softmax}(z_1), \quad p_2 = \text{softmax}(z_2)$$

Compute the ensemble probability by averaging:

$$p_{\text{ensemble}} = \frac{1}{2}(p_1 + p_2)$$

Determine the predicted class label as the index of the maximum ensemble probability:

$$\hat{y}_i = \arg \max_j p_{\text{ensemble},j}$$

return $\hat{\mathcal{Y}}$

3.7. Understanding Model Decision Making via GRAD-Cam Visualization

GRAD-CAM (Gradient-weighted Class Activation Mapping) is a strong visualization tool to understand what parts of the image provide the strongest contribution to the prediction of a model [23]. GRAD-CAM creates a heatmap of the output class in relation to the final convolutional layer, which points out significant regions of an image. GRAD-CAM visualizations were employed in this work to understand the process of decision-making by the baseline classifiers, which gives a clue to the model behavior and biases.

3.8. Web App Deployment

To make the proposed model available, we have created an interactive web application. With the help of this interface, users can post pictures of cotton plants and see what the model will forecast in real-time. The deployment offers a convenient way of showing the capabilities of the models and visualizing the decision-making processes.

3.9. Evaluation Metrics

To evaluate the performance of the proposed model, the following metrics were employed [30]:

- **Accuracy:** Measures the overall correctness of predictions.
- **Precision:** The proportion of correctly predicted positive samples out of all predicted positives.
- **F1-Score:** The harmonic mean of precision and recall.
- **Area Under the ROC Curve (AUC):** Measures the model's ability to distinguish between classes. Computed as the area under the ROC curve plotting TPR vs FPR.
- **Precision-Recall AUC (PR AUC):** Computes the area under the precision-recall curve, especially useful for imbalanced datasets.

In order to further assess the robustness and generalization property of the proposed MobileTinyVit and other baselines, 5-fold cross-validation was performed on the dataset. This step can be used to evaluate the performance of the model on various train validation splits so that the reported results are not confounded by a specific data partition and this process can also give information about the stability and consistency of the model across folds.

4. RESULTS AND DISCUSSION

In this section, a detailed analysis of MobileTinyVit will be discussed in terms of cotton leaf and plant disease detection. We present the performance of the model on various measures such as accuracy, F1 score, Cohen Kappa, and 5-fold cross-validation scores. Moreover, the time of inference and Grad-CAM images are examined to determine efficiency and interpretability. These findings give a clear insight into the strength of the model, the ability to be generalized, and the possibility of useful implementation in real-world farming environments.

4.1. Testing and Cross Validation Performance

Table 3 shows the testing performance of all the tested models on the cotton leaf disease dataset. MobileViT-S was the highest-accuracy model with an accuracy of 98.72%, precision of 1.0000, and recall of 0.9722, which means that the model was highly successful with regard to identifying diseased samples correctly and also had low rates of false positives. TinyViT-5M-224 showed similar results, having an accuracy of 98.29%, balanced precision, recall, and F1-score of 0.9892 each, and showed consistent and high-performing classification across all the classes. Other transformer models such as LeViT-192, MaxViT-Tiny, and DeiT-Small achieved a marginally lower accuracy of 97.01%, 97.44%, and 97.86%, respectively. MaxViT-Tiny recorded a higher precision (1.0000) but lower recall of 0.9462, implying that the model was quite certain in its positive results, yet it could not detect some disease samples. LeViT-192 and DeiT-Small were very precise and had high recall but failed to outperform the most successful models, which is a slight tradeoff in their overall prediction ability. MobileTinyVit soft voting ensemble proposals were overall more accurate with a 99.57% performance with 100% precision, recall, F1-score, as well as sensitivity. It also demonstrated a mean positive predictive value of 0.9882 and a mean negative predictive value of 1.0000, which means that the ensemble minimized false positives and false negatives. These findings imply that using MobileViT-S and TinyViT-5M-224 with a soft voting strategy can optimally utilize the advantages of the two models as complements to each other to achieve greater robustness and high performance in cotton leaf disease classification. In order to evaluate model reliability and cross-validation generalizability, 5-fold cross-validation was conducted on all models, and the performance measures are tabulated below. The results show the consistency of the reported performance measures.

Table 3. Testing performance for all evaluated models.

Model	Accuracy	Precision	Recall	F1-Score	Mean PPV	Mean NPV	Sensitivity
MobileViT-S	98.72%	1.0000	0.9722	0.9859	0.9879	0.9956	0.9722
TinyViT	98.29%	0.9892	0.9892	0.9892	0.9834	0.9934	0.9677
LeViT-192	97.01%	0.9891	0.9785	0.9838	0.9708	0.9900	0.9615
MaxViT-Tiny	97.44%	1.0000	0.9462	0.9724	0.9734	0.9902	0.9462
DeiT-Small	97.86%	0.9787	0.9892	0.9840	0.9782	0.9928	0.9423
Proposed MobileTinyViT	99.57%	1.0000	1.0000	1.0000	0.9882	1.0000	1.0000

Table 4 shows the 5-fold cross-validation results of transformer-based models to classify cotton leaf disease. The MobileViT-S model had the best single performance as it had the highest accuracy of 0.9976 ± 0.0003 and the Cohen's Kappa value of 0.9966 ± 0.0005 , which is considered to have high reliability and consistency in inter-fold. It had ROC AUC and PR AUC values of 1.0000 ± 0.0000 and 1.0000 ± 0.0000 , respectively, which is an indication of perfect class separability. TinyViT was next, with an accuracy of 0.9971 ± 0.0007 and Kappa of 0.9960 ± 0.0009 , which proves high model generalization. On the same note, MaxViT-Tiny had an accuracy of 0.9974 ± 0.0008 and Kappa of 0.9964 ± 0.0011 with impeccable mean AUC scores. LeViT-192 had a little lower accuracy (0.9751 ± 0.0262) and Kappa (0.9652 ± 0.0368), although high ROC AUC (0.9967) and PR AUC (0.9905) indicated successful class distinction despite slight variations. DeiT-Small was also doing a great job with an accuracy of 0.9903 , Kappa of 0.9865 , and a standard deviation of 0.0016 .

The proposed MobileTinyViT ensemble surpassed the majority of individual architectures with an accuracy of 0.9974 ± 0.0007 , Kappa of 0.9964 ± 0.0009 , and a perfect mean ROC AUC and PR AUC of 1.0000 each. Its mean specificity is great (0.9977 ± 0.0008), which proves its strength and small false positive value. Second, the backbone networks incorporated into the proposed model, MobileViT-S and TinyViT, also performed better and with more consistent cross-validation, which promotes the synergy of the networks and their role in ensuring the stability and generalization of the ensemble. All in all, the results confirm the effectiveness of transformer frameworks (especially their combination) in accurate and interpretable cotton leaf disease detection. Relatively low variances with similar results for these 5 folds suggest a good stability of model, which is enabled by the efficient data augmentation and preprocessing that generates uniform but diverse training samples. This reduces variance among the folds and it leads to a good generalization performance on the cotton leaf data.

Additionally, the validation accuracy and loss curves of the baseline MobileTinyViT model are shown in Figure 6, providing insight into the model's training stability and convergence behavior across epochs. Figure 6

Table 4. 5-Fold Cross-Validation Performance Summary for All Models

Model	Accuracy	Kappa	Mean ROC AUC	Mean PR AUC	Mean Specificity
MobileViT-S	0.9976 ± 0.0003	0.9966 ± 0.0005	1.0000 ± 0.0000	1.0000 ± 0.0000	—
TinyViT-5M-224	0.9971 ± 0.0007	0.9960 ± 0.0009	0.9999 ± 0.0001	0.9999 ± 0.0001	—
LeViT-192	0.9751 ± 0.0262	0.9652 ± 0.0368	0.9967 ± 0.0051	0.9905 ± 0.0142	—
MaxViT-Tiny	0.9974 ± 0.0008	0.9964 ± 0.0011	1.0000 ± 0.0000	1.0000 ± 0.0000	—
DeiT-Small	0.9903 ± 0.0016	0.9865 ± 0.0023	0.9998 ± 0.0001	0.9995 ± 0.0002	—
Proposed MobileTinyViT	0.9974 ± 0.0007	0.9964 ± 0.0009	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9977 ± 0.0008

demonstrates validation accuracy and loss curves of the original MobileTinyViT model to demonstrate the behavior of the training and stability of the model over epochs. The figure compares the training and validation loss and accuracy of two backbone models, MobileViT-S and TinyViT, based on their performance. In the case of the MobileViT-S model, the loss curves show that the early epochs have a very steep drop, which means that the early learning stage is successful. The training and the validation loss become stable after the second epoch, but the validation loss stabilizes at approximately 0.1, and the training loss is close to zero at epoch five. This tendency indicates the existence of efficient feature learning that is not overfitting. The accuracy curves demonstrate a steep rise during the first two epochs to about 98% with a small difference between training and validation accuracy, which implies good generalization. Conversely, the TinyViT model also shows the initial reduction in training and validation loss. Nevertheless, with continued training, the validation loss also varies slightly, whereas the training loss is decreasing towards zero. This action is a sign of slight overfitting. The

accuracy curves of TinyViT depict high training accuracy of about 99%, whereas the curvature of validation accuracy varies slightly, and almost decreases after the 6th epoch, which confirms the overfitting behavior of the two backbones. All in all, both backbones have high convergence accuracy and performance as indicated by the accuracy curves, though MobileViT-S is a little more stable and generalizes after the sixth epoch. Such findings justify the steady cross-validation results with the proposed MobileTinyViT model.

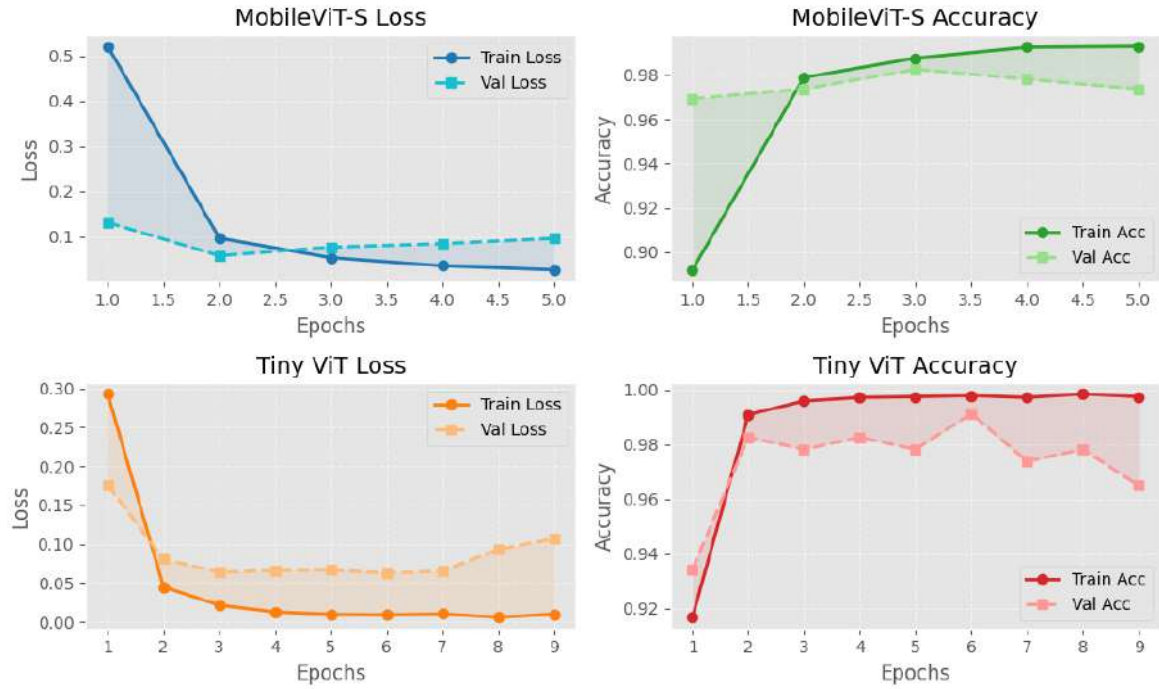


Figure 6. Training and validation performance of the baseline models used in the proposed MobileTinyViT framework. The top row shows the loss and accuracy curves for MobileViT-S, while the bottom row illustrates those for TinyViT-5M.

4.2. Quantitative Performance Evaluation

Table 5 presents a comparison between the performance of the tested transformer-based models with the help of the Matthews Correlation Coefficient (MCC), Cohen Kappa, PR AUC, and ROC AUC. Of all the models, the proposed MobileTinyViT scored highest in all metrics, with an MCC of 0.9939, a Cohen Kappa of 0.9940 and perfect PR AUC and ROC AUC values of 1.0000, demonstrating near-perfect classification performance. MobileViT-S and TinyViT-5M also performed well, with MCC values of 0.9822 and 0.9763, and PR AUC and ROC AUC values near one, indicating a high level of discriminative power. MaxViT-Tiny, LeViT-192, and DeiT-Small had lower but consistent results, which ensured their stability in a wide variety of architectures. In general, these results highlight that MobileViT-S and TinyViT-5M ensemble fusion is better in predictive accuracy with increased reliability and generalization to classify cotton leaf disease.

Table 5. Performance comparison of all models based on MCC, Cohen's Kappa, PR AUC, and ROC AUC.

Model	MCC	Cohen's Kappa	PR AUC	ROC AUC
MobileViT-S	0.9822	0.9821	0.9996	0.9998
TinyViT-5M	0.9763	0.9762	0.9975	0.9987
LeViT-192	0.9587	0.9583	0.9963	0.9984
MaxViT-Tiny	0.9651	0.9644	0.9935	0.9970
DeiT-Small	0.9703	0.9702	0.9955	0.9978
Proposed MobileTinyViT	0.9939	0.9940	1.0000	1.0000

4.3. Missclassification Analysis of Proposed MobileTinyViT

Figure 7 shows the confusion matrix of the proposed MobileTinyViT model illustrates a high level of classification performance in all four classes. Namely, 36 disease cotton leaf images were correctly identified, along with the 93 disease cotton plant images and the 53 fresh cotton leaf images. Out of the 52 fresh cotton plant images, one was wrongly identified as an infected cotton plant, and the remaining 51 were rightfully identified. This low misclassification is associated with an insignificant error rate, which indicates the high overall accuracy of the model and good discriminating capability to differentiate among the visually similar conditions of the cottons leaf and plant.

These findings verify that the MobileTinyViT ensemble can be considered efficient in capturing both local and global features and making solid and reliable predictions to classify cotton leaf disease, as well as being able to retain almost perfect predictions across all categories.

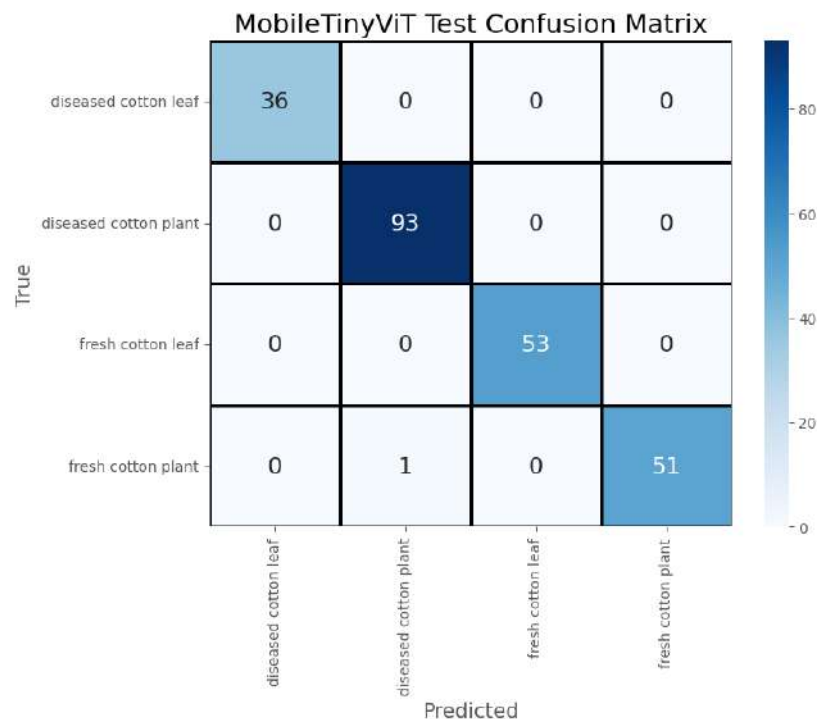


Figure 7. Confusion matrix of the proposed MobileTinyViT model for cotton leaf disease classification, showing correct predictions for all classes except one fresh cotton plant misclassified as a diseased cotton plant.

4.4. Analysis of Computational Cost

Table 6 shows the training time, inference time and the memory used by the GPU in all the models tested. LeViT-192 was the fastest training with 103.35 seconds, as well as, the lowest inference times with 9.27 seconds to train inference, 0.46 seconds to validate inference and 0.48 seconds to test inference and consumed 0.50 GB of gpu memory. On 0.10 GB of the GPU memory, MobileViT-S took 299.61 seconds to train, 13.24 seconds to infer, 0.59 seconds to validate and 0.61 seconds to test, and did this with 299.61 seconds of training time. The training time of TinyViT is 232.49 seconds; the training, validation, and testing times are 10.77 seconds, 0.51 seconds, and 0.52 seconds, respectively; and it consumed 0.13 GB of GPU memory. MaxViT-Tiny had a training inference of 211.03 seconds, and a higher training inference of 31.68 seconds, both validation and test inferences of 1.20 seconds and a 0.52 GB GPU memory usage. DeiT-Small took 276.94 seconds to train, inference times of 15.52 seconds to train, 0.67 seconds to validate, and 0.69 seconds to test and consumed 0.71 GB of GPU memory.

On the other hand, The proposed MobileTinyViT ensemble required the longest training time at 464.44 seconds due to the combination of two backbone models but it possesses less parameters (approximately 7.2M in total) than heavier hybrids for saving GPU memory usage remained below 1 GB (0.77 GB), and its inference

times were negligible at 15.52 seconds for training inference, 0.67 seconds for validation, and 0.56 seconds for testing. This helps in its real time application for deployment over the web and mobile platforms. Moreover, soft-voting ensemble enhances trustworthiness by decreasing single model error and also produces 1% higher accuracy than single lightweight models. These results indicate that the ensemble offers a practical balance between high accuracy, rapid inference, and efficient GPU utilization, making it feasible for real-time cotton leaf disease detection despite longer training time. .

Table 6. Training time, inference time, and GPU memory usage for all models.

Model	Train Time (s)	Train Inf (s)	Val Inf (s)	Test Inf (s)	GPU (GB)
MobileVit_s	299.61	13.24	0.59	0.61	0.10
TinyVit	232.49	10.77	0.51	0.52	0.13
LeViT-192	103.35	9.27	0.46	0.48	0.50
MaxViT-Tiny	211.03	31.68	1.20	1.20	0.52
DeiT-Small	276.94	15.52	0.67	0.69	0.71
Proposed MobileTinyVit	464.44	15.52	0.67	0.56	0.77

4.5. Understanding Model Decision Making Process via GRAD-Cam

Figure 8 displays the Grad-CAM images of the proposed MobileTinyVit model and it gives an idea of how the model attends to the relevant areas during prediction. The upper row illustrates the diseased cotton leaf samples, whereas the lower row illustrates the diseased plant samples. The heatmaps show that the model continuously focuses on the diseased areas of the plants and leaves, like lesions, spots, and discoloration, but mostly neglects the unrelated parts not within the diseased areas. This illustrates that the model is successful in identifying and making use of discriminative characteristics associated with the disease.

The Grad-CAM outputs were verified by expert annotations and the regions identified by the model are close to the ones addressed by the specialists. Such findings confirm that the attention mechanism of the model is oriented on the disease-specific regions in a correct manner, which gives interpretability to the decision-making process and trust to the decision. Altogether, the grad-cam analysis suggests that the suggested MobileTinyVit model is highly dependable in detecting cotton leaf disease through the focus on clinically significant areas of the leaf picture, as well as the plant picture.

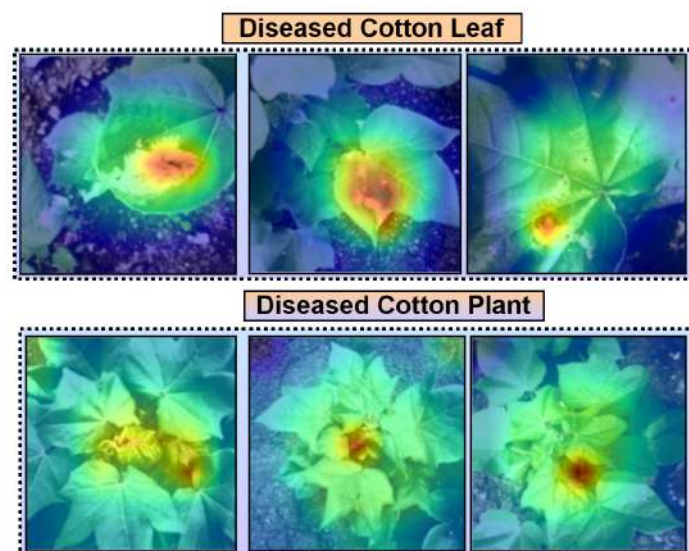


Figure 8. Grad-CAM visualization of the proposed MobileTinyVit model. The figure shows both diseased cotton leaf (top) and plant (bottom) samples, highlighting the regions the model focuses on for disease detection.

4.6. Prediction via Web App

The figure 9 demonstrates the predictions of diseased cotton leaves and plants with the help of the developed web application of the real-time prediction. Users are allowed to feed the app with images of leaves or plants, which is then processed by the trained model and gives a prediction in 2 seconds.

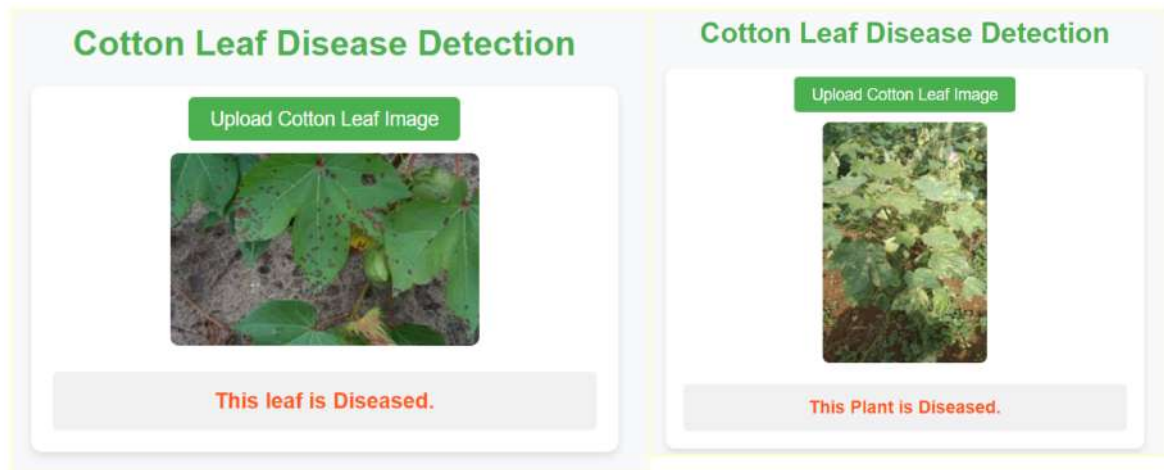


Figure 9. Web application for real-time prediction. The system allows users to input data/images and obtain model predictions efficiently.

The visualization proves that the system effectively detects disease-affected regions, allowing its quick evaluation and decision-making. This underscores the practicality of the proposed model in the real-time user-friendly context.

4.7. Comparative Analysis with previous studies

A comparative analysis of the recent research on cotton leaf disease detection is discussed in Table 7 and summarizes the type of models used, the reported accuracies, and limitations. Past literature has shown good performance with convolutional neural networks, fine-tuned deep-learning models, hybrid models, and ensemble-based models with accuracies of roughly 97–98.7

Conversely, The MobileTinyViT attains 99.57% accuracy, a slight lead over the recent works (98.5–98.7%) on similar tasks by themselves fairly impressive results for small models and challenging testing setups. This gain is enabled by effective class balance aware data augmentation, robust pre-processing and soft voting ensembles that reduce error of individual models. While it is hard to say for sure because of the differences in dataset and conditions, the practical advantage can be found considering high performance with low inference time (0.56 s), and its interpretability of Grad-CAM will definitely make a difference in real-time deployment.. Comprehensively, the findings show that MobileTinyViT not only enhances the accuracy of the classification but also offers a holistic end-to-end solution to perform fast and accurate cotton disease management.

Table 7. Comparison of Recent Cotton Leaf Disease Detection Studies

Study	Year	Model	Accuracy (%)	Limitations
[15]	2025	CNN	98.7	Relies on high-quality images, requires powerful hardware, potential overfitting, uncertain performance in real-field conditions
[10]	2023	Xception (fine-tuned CNN)	98.7	Class imbalance, lack of explicit feature segmentation, limited real-time deployment integration
[17]	2025	BERT-ResNet-PSO hybrid	98.5	Increased model complexity, high computational cost, limited validation on unseen environments
[16]	2025	EfficientNetB3 + Inception-ResNetV2	98.0	Generalization to diverse real-world conditions untested, challenges in real-time deployment
[8]	2025	StackNet (CNN + ML classifiers + GAN)	97.0	GAN-generated artifacts, limited interpretability, increased computational cost
Our Study	2025	MobileTinyVit (MobileViT S + TinyViT 5M 224)	99.57	Strength: Lightweight, fast inference, interpretable with Grad-CAM, suitable for real-time deployment

5. CONCLUSION AND FUTURE WORK

This paper introduces MobileTinyVit, which is a lightweight soft voting ensemble of MobileViT S and TinyViT 5M 224, to detect cotton leaf and plant diseases effectively and efficiently. The proposed model shows exemplary performance with an accuracy of 99.57%, F1 score of 1.0000, Cohen Kappa of 0.9940, and a stable 5 fold cross validation score of 0.9974 ± 0.0007 . Grad CAM visualizations confirm that the model is effective in targeting disease specific regions, which increase interpretability. The suggested model can also be inferred faster in 0.56 seconds with the minimum use of the GPU, and the deployed web application can make live predictions in less than 2 seconds, which proves its feasibility in precision agriculture. One limitation of this study is that it does not target or distinguish specific leaf or plant disease category.

The model has been tested on a single dataset with more constrained imaging environment, corresponding to overfitting and reduced ability for generalization to various real-field scenarios. The method does binary classification (Diseased or healthy) and does not recognize the specific type of disease, so it is difficult to diagnose and treat specifically. The model design is specific for cotton plants, and therefore cannot be directly used for other plants without retraining. While some of these variations are addressed directly by augmentation, the robustness to extreme conditions (i.e., heavy occlusion, adverse weather and temporal progression) and real-world farmer deployment still needs to be tested.

To improve the model further in the future, it can be generalized to manage several crops and a wider spectrum of diseases. Multimodal data of other factors like environmental conditions or soil parameters, might also enhance the accuracy of the prediction. Also, it would be possible to investigate the use of edge devices and their combination with autonomous farming systems to achieve complete disease monitoring. Lastly, implementing MobileTinyVit alongside explainable AI architecture might increase the level of trust and adoption by agricultural practitioners.

DATA AVAILABILITY STATEMENT

The dataset used in this study is publicly available at <https://www.kaggle.com/datasets/janmejybhoy/cotton-disease-dataset>.

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